EIGSEP Memo 009

Marjum Pass 2026-07: what the night fit can and cannot separate

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Abstract

The 07-17 parked night measures two numbers per frequency: a multiplicative group θ and an additive group P. Everything else the image-domain model contains (the kelvin scale, the antenna-side loss, the sky and beam scales, the ground temperature, the receiver's noise waves and a remainder) is a split of one of them, set by a prior or by an independent measurement. This memo shows each split as its posterior, so that the trade-off can be seen, states what each split rests on, and names what would break it, including what the next deployment must change. The night fit itself is memo 008's (§ 5.1); the calibration is `tcal@v0004` (memo 004). Every product here is diagnostic. Every figure, table and quoted number is regenerated by `make_products.py` in this directory from the inputs in its `inputs/` directory (Appendix A).

1 Summary

- **θ and P are measured jointly.** Their correlation is $\rho = -0.9936$ to -0.9979 at all 43 frequencies. It is not a degeneracy of the data: P is the level extrapolated to zero sky, far from the night's sky. The night's mean level $M = \theta\bar{A} + P$ is known to 0.2–3.0 K, against $\sigma_P = 2.8$–36.7 K (§3.1).
- **The bench kelvin scale and the loss prior disagree.** The loss prior on the antenna-side gain G, built from the parts' datasheets, gives $\theta/G = 1.62 \pm 0.03$; the bench measurement of the noise-source temperature gives $k_{\mathrm{bench}} = 1.15$. The tension is $+6.0$, $+3.7$, $+3.0$ and $+1.9\,\sigma$ from low to high band with GSM2016, and $+3.2$, $+1.5$, $+0.9$ and $+0.7\,\sigma$ with McKay et al.'s sky (§3.2).
- **The datasheets cannot supply the G the bench needs.** If the bench k is right, $G = 1.07$, 1.05, 0.96 and 1.04 by band, consistent with a lossless path but above the measured switch path's own G_{sw} at all 34 channels with a bench k (by 0.3–2.5 σ). The difference therefore lies in the sky scale, the beam normalisation or the bench k, unless the coax and balun are far off their datasheets. An in-situ check of the loss, limited by the antenna model, neither confirms nor overturns the prior (§3.2.1).
- **The loss prior's width is an allowance.** The allowance $\Delta \sim U[0, 0.2]$ dB for the balun board and the transformer's unit spread carries 97–99 % of the insertion loss's prior variance, so the width of G is set by judgement, not measured. It sets the width of $k = \theta/G$ up to 101.56 MHz; from 183.59 MHz up, θ's jackknife error is the larger (§3.2.1).
- **If the bench k and the loss prior both hold,** the sky-times-beam scale is $s = 1.3$–1.7 (band means; McKay et al.'s sky changes θ by only 16–21 %) (§4).
- **Inside P, the ground-temperature prior decides.** The prior $T_{\mathrm{gnd}} \sim \mathcal{N}(293, 30)$ K sets 58–96 % of the variance of the additive remainder T_{off} (band means): a T_{off} or T_{gnd} quoted from this night is the prior's (§3.3).
- **The receiver's noise waves are an unresolved systematic in P.** The bench noise-source temperature does not pin them: T_{unc} is -14 to $+21$ K up to 221.9 MHz, inside its 22–33 K error, T_0 is negative above, and no solution passes the held-out test. Mapped into P, the noise-wave prior would shift band means by up to $+23.3 \pm 5.3$ K (± 29 K with the prior's covariance inflated to its held-out residual), beyond P's joint error in 3 of four bands. The shift comes from the phase noise waves, which the data do not support (shape test $w = -0.10 \pm 0.07$, against 1 predicted); it is not in P's quoted error and does not touch θ (§3.4).

- **Frame.** All temperatures are in `tcal@v0004`'s frame. The next calibration release scales temperature differences from the reference load's by $r = 0.85$–0.88 (§2).

2 The model and its groups

The input is memo 008's night fit at the measured tilt (§ 5.1 there): the 07-17 parked night (04:12–11:24 UTC), 318 of its 362 bins, 43 frequencies from 50.78 to 234.38 MHz, `fit_v0025`'s calibrated data, with the box at the axle tilt box-air's IMU measures (memo 003). It fits each frequency separately:

$$y(t) = \theta\, A(t) + P + (1-k)\,[c_L(t) - \bar{c}_L], \qquad A(t) = S(t) + T_{\rm CMB}\,[1 - f(t)], \tag{1}$$

with y the calibrated spectrum at the receiver's reference plane, S the beam-weighted sky (GSM2016; Zheng et al. 2017), f the beam's ground fraction, and c_L the load term, which follows from the calibration equation and enters as a fixed column, iterated with the fit, with $k = \theta/\bar{G}$ and $\bar{G}$ the loss prior's median G ($1-k$ from -1.10 to -0.46). θ and P therefore depend on the loss prior through this column. It is kept because it moves θ or P by more than σ_J at 4 frequencies, a change larger than the reported error; it raises the held-out χ^2 per bin from 27.17 to 27.50 on the same test bins. The fit returns θ and P with a delete-one-block jackknife covariance over 15 blocks of 30 min.

Selection. The box moves under wind load on its ropes with its motors still, and a moving box's beam is not where the model puts it. The fit drops every bin in a 5-min window in which the IMU's per-sample tilt has a standard deviation above 0.2° or a sample flagged as not quasi-static: 7 windows, 44 of the 362 bins. The early windows coincide with the end of the night's strong S–SW wind; the later ones fall in light E–SE wind, no stronger at the nearest stations than in the calm hours (memo 008, § 5.1). The rule is defined on the instrument and was fixed before it was scored; on the test bins it leaves, it lowers the held-out χ^2 per bin from 30.02 to 27.50 (lower at 34 of 43 frequencies).

Errors across frequency. The jackknife replicates delete the same 30 min at every frequency, so the frequencies' errors are strongly correlated. A mean over frequency (all-band or by band) takes its error from the spread of the replicates' means, never from the per-frequency errors combined as if independent: θ's all-band mean is 1.169 ± 0.012, against ± 0.003 if the frequencies were independent. Every prior or measurement that is one physical quantity enters a band mean as one draw for all frequencies: the bench's kelvin scale (its shape measured, its level uncertain), the ground temperature and the outdoor air temperature.

The groups. The two groups hold these pieces:

$$\theta = k\,G \times (\text{sky scale}) \times (\text{beam normalisation}), \tag{2}$$

$$P = kG\,\bar{f}\,T_{\rm gnd} + k\,\bar{T}_{\rm add} + (1-k)\,\bar{c}_L + T_{\rm off}. \tag{3}$$

Here $k = T_{\rm NS}(\texttt{tcal@v0004})/T_{\rm NS}(\text{true})$ is the kelvin scale: the noise-source temperature `tcal@v0004` carries over the true one, so $k > 1$ if the true $T_{\rm NS}$ is lower. $G = G_{\rm sw}G_{\rm cb}$ is the available gain of the antenna-side path: the switch path $G_{\rm sw}$, measured, and the balun and coax $G_{\rm cb}$. The loss prior on G builds $G_{\rm cb}$ from the parts' datasheets: the balun's transformer (ADT1.5-1+), 1.32 m of RG-401 coax, and an allowance for the balun board and the transformer's unit spread (§3.2.1). $T_{\rm add} = (1-G_{\rm sw})\,T_{\rm sw} + G_{\rm sw}(1-G_{\rm cb})\,T_{\rm out}$ is the lossy path's own emission, with $T_{\rm sw}$ the switch's logged temperature and $T_{\rm out}$ the outdoor air temperature, given a prior (§3.3). c_L is the reference load's mismatch-weighted temperature and $\bar{c}_L$ its night mean; $\bar{f}$ is the night's mean ground fraction, 0.47–0.67 across the band; $T_{\rm off}$ is the additive term that remains. The ground term is $kG\,\bar{f}\,T_{\rm gnd}$: the beam is normalised over the full sphere, so a sky-times-beam scale does not reach the ground.

$T_{\rm off}$ is not the receiver's offset, which cancels in `tcal@v0004`'s three-state calibration. It holds kD and $T_{\rm rest} = T_{\rm off} - kD$, whatever additive term is left. D is the difference between the receiver's noise waves on the antenna and on the load (Monsalve et al., 2017):

$$D = \sum_{i \in \{\rm unc,cos,sin\}} c_i\,T_i, \qquad c_i = -X_{a,i} + \frac{M_L}{M_a}\,X_{L,i}, \tag{4}$$

Table 1: The night fit's degenerate groups and where this memo treats them.

Group	What is in it	What breaks it	Status
θ against P	the multiplicative and additive groups	the sky's drift through the night (data)	measured jointly, correlated (§3.1)
k against G (in θ)	kelvin scale, antenna-side gain, sky scale, beam normalisation	bench k; the coax and balun measured; sky scale (GSM2016 against McKay et al.); raster beam coverage	**in tension** (§3.2)
T_{gnd} against T_{off} (in P)	ground emission, the remaining additive term (kD and a remainder)	a ground-fraction lever (heights, rotation); T_{gnd} measured; T_{off} from the noise waves	prior-determined (§3.3)
noise waves against P	T_{off}'s receiver part	field calibrators with the bench T_{NS}, mapped into P	candidate breaker, not adopted (§3.4)
beam against sky	the shape of $S(t)$	the transmitter raster	open (§3.5)
tilt against a time trend	axle tilt, generic time terms	the IMU	broken (§3.5)
load term against the scale	$(1-k)\,c_L(t)$	fixed column, iterated	handled (§3.5)
gain drift against the sky	a time-varying receiver gain $g(t)$ against the sky's time dependence	the switch cycles as a gain series	open (§3.5)

with T_{unc} the receiver's uncorrelated noise wave, T_{cos} and T_{sin} its phase noise waves, $X_{a,i}$ and $X_{L,i}$ the noise-wave coefficients of the antenna and of the load (Monsalve et al.'s Eq. 7, functions of their reflection coefficients and the receiver's), and $M = 1/X_L$. The receiver's own noise offset T_0 cancels with the load. T_{rest} has no model of its own: it is a leftover, so no split of P is a measurement until it is bounded. The posteriors of §3 take the sky scale and the beam normalisation as 1, so $\theta = kG$ there; §3.2 returns to them.

The kelvin frame. Every temperature here is in `tcal@v0004`'s frame, whose noise-source temperature is `noise_source@v0001`'s. The bench measurement of the noise-source temperature, with the hot load's path through the RF switch, puts it at $r = 1/k_{\mathrm{bench}}$ times v0001's: $r = 0.876/0.877/0.858/0.863$ on average over 50–88, 108–150, 150–200 and 200–235 MHz, and 0.85–0.88 at the 34 frequencies with a bench k, as in memo 008 (§ 5.1). The next calibration release (`tcal@v0005`, prepared, not yet released) carries the bench temperature. Every temperature difference from the reference load's then becomes r times its value here, and θ and k with it; the bench k becomes 1. Absolute temperatures such as T_{gnd} do not scale. The rescale multiplies θ/G and the bench k alike, so the tensions of §3.2 do not change.

The fit against the data. Memo 008 (§ 5.1, Figs. 4 and 5) shows the fit against the data. With the IMU's tilt the residual is 2.5–12.4 times the bin noise σ_{tot} (rms over the night, per frequency; worst at 226.56 MHz), against 5.6–18.9 with the axle tilt fitted with `empirical_raster_v0017` ($\tau = 4.74°$). It is structured in time, so the model of Eq. (1) does not describe the night to the noise. Its cause is open; every split below is conditional on Eq. (1).

Figure 1 gives the fitted parameters against frequency. θ runs from 0.83 to 1.44 (weighted mean 1.169 ± 0.012, the error joint across frequency), with a jackknife error of 0.9–17 % (worst at 226.56 MHz). P has weighted mean 145.8 ± 3.6 K with errors of 2.8–36.7 K (worst at 50.78 MHz). P's error does not include the receiver's noise waves, an unresolved systematic of up to tens of kelvin in a band mean (§3.4); θ is not affected. The pointing, horizon and tilt variants move θ by 1.3 % and P by 3.8 K (weighted means of the band's half-width), at most 1.72 and 1.41 times the jackknife errors.

Table 1 lists the groups. Each split value carries what set it: a measurement or a prior. No group is split by the night fit alone.

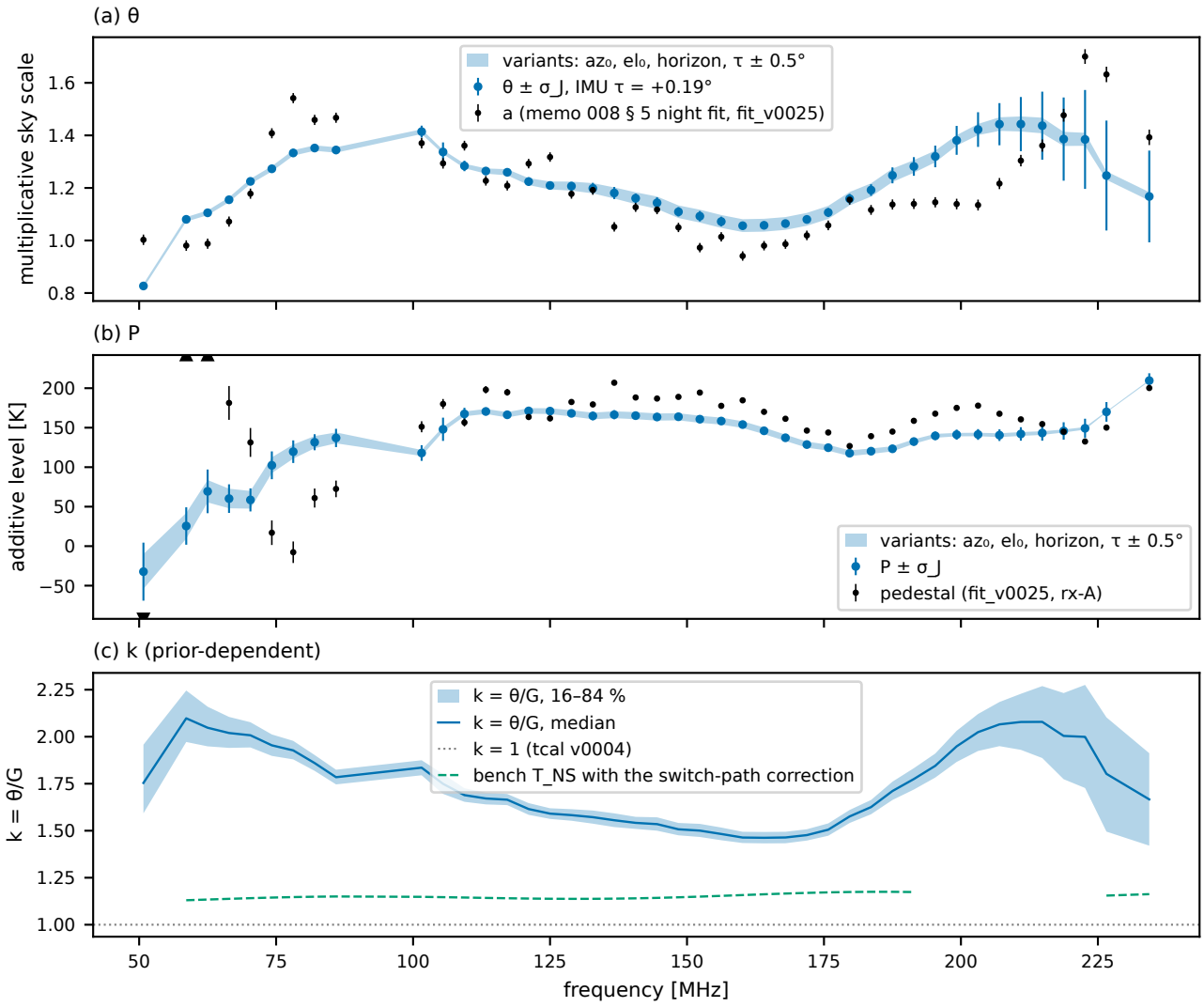


Figure 1: **The fitted groups against frequency.** The night fit (Eq. 1) at the tilt it holds, $\tau = +0.19°$, the IMU's night mean as the fit took it (§3.5; blue, $\pm\sigma_J$ the jackknife error), memo 008 § 5's night fit `fit_v0025` (black), and the band of the pointing, horizon and tilt ($\tau \pm 0.5°$) variants (light blue). (a) θ, the multiplicative sky scale, against `fit_v0025`'s a; (b) P, the additive level, against its rx-A pedestal (triangles: points beyond the panel); (c) the after-fit split $k = \theta/G$ over the loss prior (16–84 %), with $k = 1$ (`tcal@v0004`, dotted) and the bench T_{NS} with the switch-path correction (dashed). Panel (c) is a prior-determined split; its width is mostly the loss prior's allowance up to 101.56 MHz and mostly θ's jackknife error from 183.59 MHz up (§3.2.1); panels (a) and (b) are the data; the correlation of θ and P is in §3.1. Memo 008's Fig. 7 shows the same fit. Diagnostic.

3 The pairs

3.1 θ against P: correlated, because P is the level at $A = 0$

Over a parked night $A(t)$ changes only by the sky's drift, whose rms is 8–11 % of its mean $\bar{A}$. The data therefore fix the mean power $\theta\bar{A} + P$ well, and the split between θ and P only through A's variation. The correlated direction is $\Delta P = -\bar{A}\,\Delta\theta$. This is not a degeneracy of the data: P is the level extrapolated to zero sky, $A = 0$, far from the night's values of A.

Figure 2 shows the posterior. The correlation is $\rho = -0.9936$ to -0.9979 at every frequency, with band means -0.996, -0.995, -0.997 and -0.996 (50–88, 108–150, 150–200 and 200–235 MHz). The posterior's long axis lies along $-\bar{A}$ to within a factor 0.94–1.07. At 70.31, 148.44 and 199.22 MHz the mean power is known to 1.5, 0.4 and 0.5 K, against $\sigma_P = 14.6$, 4.2 and 6.4 K.

Reported instead as θ and the mean level $M = \theta\bar{A} + P$ ($\bar{A}$ here the $\sigma_{\rm tot}$-weighted mean of A over the fitted bins; Fig. 2), the night measures M to 0.2–3.0 K (mean 0.9 K) against P's 2.8–36.7 K. θ and M are uncorrelated in the fit's formal covariance, but the block jackknife gives $\rho(\theta, M) = -0.72$ to 0.61 (mean -0.02; most negative at 156.25 MHz) because the residual is structured (§2). M is therefore much better determined than P, but not independent of θ while the residual is structured.

Measured: θ and P jointly, from the data, with no prior. **Breaker:** more variation of A at one kelvin scale, for example the same receiver at other sidereal times. **Status:** measured; the split is the night's limit.

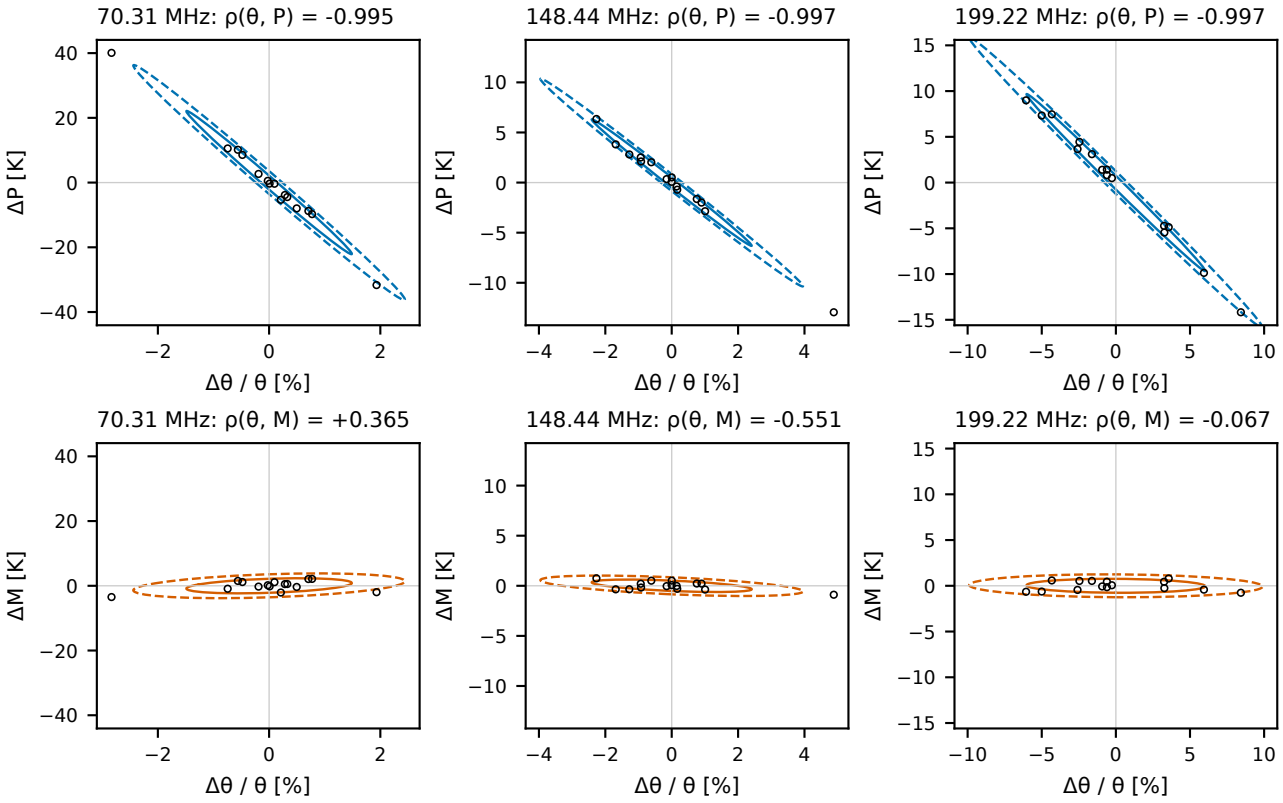


Figure 2: θ **against** P **and against** M**.** The night fit at 70.31, 148.44 and 199.22 MHz, as offsets from the fit: $(\Delta\theta/\theta, \Delta P)$ (upper, blue) and $(\Delta\theta/\theta, \Delta M)$ with $M = \theta\bar{A} + P$ (lower, vermillion). Circles: the 15 delete-one-block replicates, scaled by $\sqrt{n-1}$ so that they scatter as the jackknife covariance C_J; lines: C_J's 68 % (solid) and 95 % (dashed) contours; titles: the jackknife correlation. Each column shares one vertical scale. (θ, P) lies along $\Delta P = -\bar{A}\,\Delta\theta$, the direction that leaves M unchanged; M is known far better than P. No prior enters. Diagnostic.

Table 2: The adopted loss prior's pieces (Eq. 5) at the channels nearest 50, 70, 100, 150 and 200 MHz, at the night fit's antenna sweep. G_{sw}: the switch path's available gain from its measured S-parameters. ADT1.5-1+: the datasheet's typical insertion loss (IL). Balun board: an upper bound from its production files. RG-401: the range of α_{100}. Lower block: the prior's draws, median with its 16th and 84th percentiles. Figure 3 shows every row against frequency; the in-situ estimate is in Fig. 4.

frequency (MHz)	50.78	70.31	101.56	148.44	199.22
switch path G_{sw} (measured)	0.876	0.794	0.906	0.880	0.871
ADT1.5-1+ IL (dB, typical)	0.40	0.42	0.47	0.53	0.61
balun board IL (dB)	$\lesssim$ 0.05–0.07 at every frequency (production files)				
RG-401, 1.32 m, IL (dB)	0.056–0.073	0.066–0.086	0.080–0.104	0.096–0.125	0.112–0.145
Δ (dB)	uniform on [0, 0.2]				
total IL (dB)	$0.56^{+0.07}_{-0.06}$	$0.60^{+0.07}_{-0.07}$	$0.66^{+0.07}_{-0.07}$	$0.74^{+0.07}_{-0.06}$	$0.84^{+0.07}_{-0.07}$
$L = 10^{-\mathrm{IL}/10}$	$0.878^{+0.014}_{-0.014}$	$0.871^{+0.014}_{-0.013}$	$0.860^{+0.013}_{-0.014}$	$0.843^{+0.013}_{-0.014}$	$0.824^{+0.013}_{-0.013}$
G_{cb}	$0.538^{+0.054}_{-0.056}$	$0.768^{+0.026}_{-0.025}$	$0.851^{+0.014}_{-0.015}$	$0.836^{+0.014}_{-0.014}$	$0.814^{+0.014}_{-0.014}$
$G = G_{\mathrm{sw}}G_{\mathrm{cb}}$	$0.471^{+0.048}_{-0.049}$	$0.610^{+0.020}_{-0.020}$	$0.771^{+0.013}_{-0.013}$	$0.735^{+0.013}_{-0.012}$	$0.709^{+0.012}_{-0.012}$

3.2 k against G: the bench scale and the loss prior

The night measures θ only. Each draw of the loss prior on G (§3.2.1) gives $k = \theta/G$, so the posterior lies along the hyperbola $kG = \theta$, with its length set by the G prior (Fig. 5). The independent measurement is the bench k in `tcal@v0004`'s frame, $k_{\mathrm{bench}} = 1.13$–$1.17$ (mean 1.15) with a 10.4–10.8 % error, mostly the assumed transfer from bench to field, which is taken as one scale error for all frequencies. The bench's hot load went through the RF switch, as the bench's switch-path correction assumes; the bench measurement masks 8 channels from 195.31 to 222.66 MHz.

3.2.1 The loss prior and its evidence

The prior builds the antenna-side path from its parts:

$$\mathrm{IL} = \mathrm{IL}_{\mathrm{ADT}}(\nu) + l\,\alpha_{100}\sqrt{\nu/100\,\mathrm{MHz}} + \Delta, \quad L = 10^{-\mathrm{IL}/10}, \quad G_{\mathrm{cb}} = L\,\frac{1-|\Gamma_X|^2/L^2}{1-|\Gamma_X|^2}, \quad G = G_{\mathrm{sw}}G_{\mathrm{cb}}, \tag{5}$$

with $\mathrm{IL}_{\mathrm{ADT}}$ the balun transformer's typical insertion loss (Mini-Circuits ADT1.5-1+ datasheet, linear in ν between its points), $l = 1.32$ m of RG-401 coax, $\alpha_{100} \sim U[0.060, 0.078]$ dB/m (from the typical loss to the datasheet maximum, scaled from 1 GHz as $\sqrt{\nu}$), $\Delta \sim U[0, 0.2]$ dB, and Γ_X the antenna's reflection coefficient at the switch's antenna port (the measured one, de-embedded through the switch path). L is the coax and balun's one-way power transmission; G_{cb}, their available gain, falls below L where the antenna is badly matched, which widens it at the lowest frequencies. Table 2 gives each piece at five channels.

The transformer dominates the loss at every channel. The balun board, read from its production files (a short microstrip on thin FR-4, and an SMA jack), adds at most 0.05–0.07 dB. Δ covers the board and the transformer's unit-to-unit spread, which the datasheet does not bound: it gives typical insertion loss only. With Δ on $[0, 0.1]$ dB (the board alone) the low band's median G rises from 0.615 to 0.632. The prior this replaces had a coax of unknown type and length, $l \sim U[0.3, 2]$ m and $\alpha_{100} \sim U[0.15, 0.45]$ dB/m, and $\Delta \sim U[0, 0.5]$ dB. Against it, the adopted prior raises G's median by 10–23 % by band and narrows its 16–84 % range 2.9–4.1 times (Table 3).

The prior's width is mostly the allowance. The transformer enters at its typical loss with no spread, and the coax's α_{100} range gives a standard deviation of 0.007 dB at 100 MHz, against 0.058 dB for Δ; Δ therefore carries 97–99 % of the insertion loss's prior variance across the band. Of Δ's 0.2 dB, the board's bound accounts for at most 0.07 dB; the rest is a judgement allowance for the transformer's unit spread. The width of G (Fig. 5, lower left) is therefore set by that allowance, not measured. It is the larger part of the 16–84 % band of $k = \theta/G$ (Fig. 1c) up to 101.56 MHz; from 183.59 MHz up, θ's jackknife error is larger than G's relative width.

Figure 3 shows the pieces and the totals against frequency, with the G they give against the G the bench k needs. Its main conclusion does not depend on Δ: no passive coax and balun let G exceed G_{sw}, whatever their loss.

Table 3: The loss prior by band: plain means over each band's channels of the per-channel median and 16th and 84th percentiles, for the adopted prior and the one it replaces (Fig. 3 shows them against frequency). Table 4's G_{prior} weights the channels by the prior's width, so it differs from these means.

	50–88	108–150	150–200	200–235 MHz
L, adopted prior	$0.871^{+0.014}_{-0.013}$	$0.850^{+0.013}_{-0.014}$	$0.833^{+0.013}_{-0.013}$	$0.816^{+0.013}_{-0.012}$
G, adopted prior	$0.615^{+0.024}_{-0.025}$	$0.755^{+0.012}_{-0.012}$	$0.726^{+0.012}_{-0.013}$	$0.696^{+0.011}_{-0.012}$
L, previous prior	$0.804^{+0.041}_{-0.040}$	$0.772^{+0.044}_{-0.048}$	$0.750^{+0.047}_{-0.052}$	$0.730^{+0.048}_{-0.057}$
G, previous prior	$0.498^{+0.072}_{-0.069}$	$0.685^{+0.040}_{-0.044}$	$0.647^{+0.044}_{-0.050}$	$0.615^{+0.045}_{-0.052}$
L, in situ (50–77 MHz; Fig. 4)	0.965 ± 0.017	–	–	–

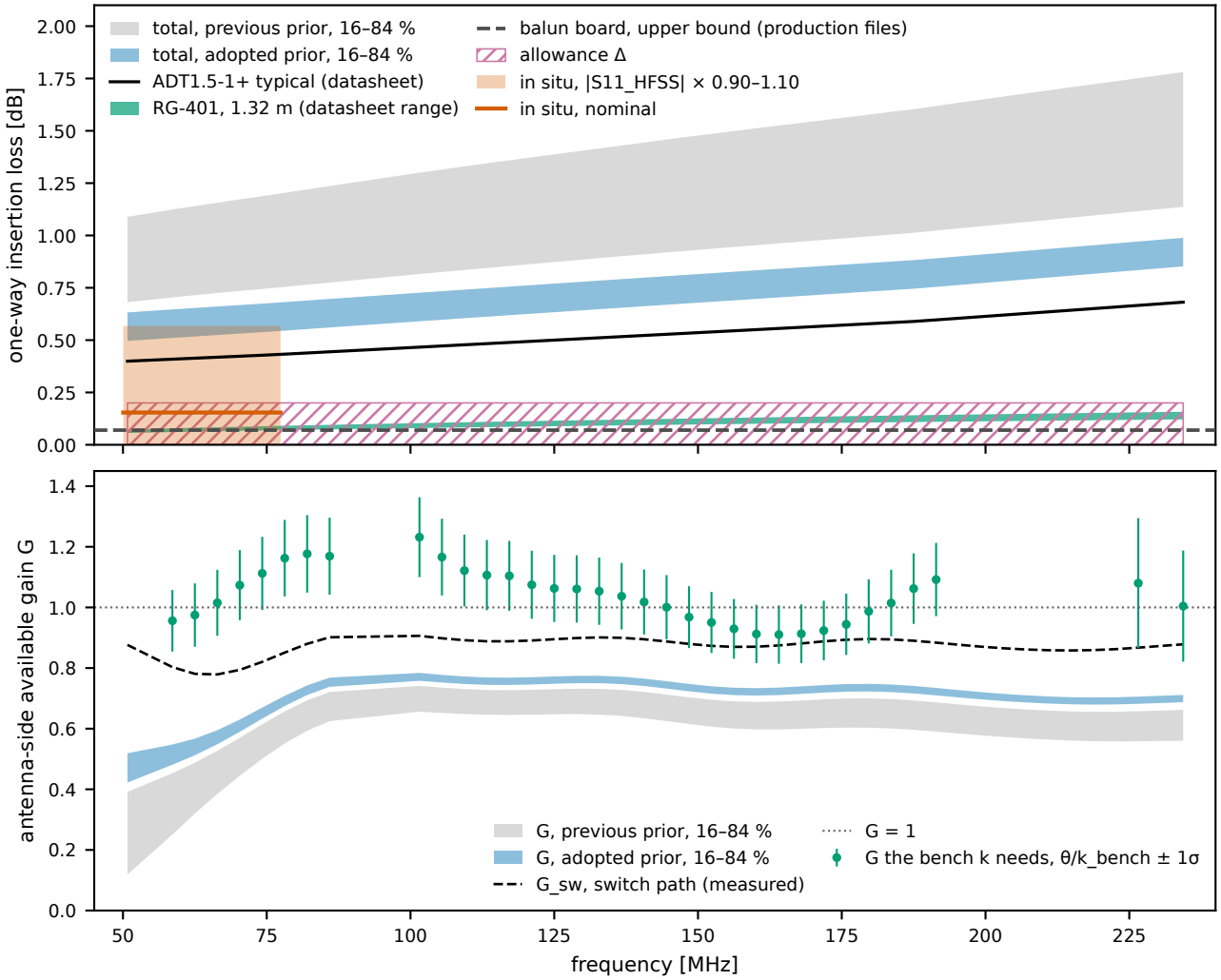


Figure 3: **The loss prior's evidence.** Upper: the one-way insertion loss of the coax and balun against frequency. Pieces: the ADT1.5-1+ transformer's typical loss (black, datasheet), 1.32 m of RG-401 between its typical and maximum loss (green), the balun board's upper bound from its production files (dark grey dashed), and the allowance Δ (purple hatched). Totals: the adopted prior (blue) and the prior it replaces (grey), 16–84 % of their draws. Vermillion: the in-situ estimate over its fit range (50–77 MHz; band: HFSS's $|S_{11}|$ scaled by 0.90–1.10, Fig. 4). Lower: the antenna-side available gain G the two priors give (Eq. 5), the measured switch path's G_{sw} (dashed), which caps G for any passive coax and balun, and the $G = \theta/k_{\text{bench}}$ that the bench k needs ($\pm 1\sigma$ from the bench's scale error and θ's jackknife; none where the bench is masked). The adopted prior moves G toward the bench but stays far below it, and the bench's G lies above G_{sw} everywhere. Diagnostic.

The in-situ check (Fig. 4). Where the antenna reflects most of its power, the reflection measured behind the coax and balun is the antenna's own, attenuated by the line on the way in and out. For a matched line, $|\Gamma_X| = L\,|\Gamma_t|$, with Γ_t the antenna terminals' reflection coefficient, so with a trustworthy model of the antenna's $|S_{11}|$ the ratio $R = |\Gamma_X|/|S_{11,\mathrm{HFSS}}|$ measures the loss in situ. Over the 113 channels with $|S_{11,\mathrm{HFSS}}| \geq 0.4$ (50–77 MHz), $L = 0.965 \pm 0.017$ (band mean, block jackknife), against the adopted prior's 0.874 (0.860–0.887) on the same channels; the 16 antenna sweeps agree to 0.003 in band-mean L. It cannot be used yet:

- A level error in HFSS's $|S_{11}|$ is fully degenerate with L: with $|S_{11,\mathrm{HFSS}}|$ 10 % higher, the estimate is 0.877, inside the prior.
- R is above the datasheet floor (the transformer's typical loss plus the coax at its typical loss, no allowance) by more than the bound on the line's mismatch ripple at 64 of the 113 channels (57 %, against a limit of 20 % fixed beforehand), and above 1 at 25 %, which a passive line cannot do.
- R rises nearly linearly, from 0.90 to 1.03, across 50–77 MHz. A loss does not have that shape (a passive line's L falls with frequency); it suggests the model's $|S_{11}|$ differs from the antenna's in shape, for example by a small offset of a resonance between the simulated and the real antenna.

The check therefore neither confirms nor overturns the adopted prior, which rests on the datasheets. It would become usable with an HFSS iteration whose $|S_{11}|$ matches the antenna as deployed to a few per cent at 50–80 MHz (the port definition and reference impedance, the conductor losses, the box and the ground beneath it), or it can be bypassed by bench measurements of the coax's S_{21} and of a back-to-back pair of baluns.

3.2.2 The tension under the adopted prior

At 70.31, 148.44 and 199.22 MHz, $k = \theta/G$ is 2.01, 1.51 and 1.95; across the band its weighted mean is 1.62 ± 0.03 (1.46 to 2.10, worst at 58.59 MHz). The tension between θ/G and k_{bench}, $z = \mathrm{mean}(\Delta k)/\mathrm{sd}(\Delta k)$ over paired draws, runs from $+1.8$ to $+6.4\,\sigma$ per channel. For a band, z is that of the band's weighted mean of Δk over draws that are joint across frequency: θ's jackknife replicates, the loss prior's frequency templates, and one bench scale for every frequency. That is $+6.0$, $+3.7$, $+3.0$ and $+1.9\,\sigma$ from low to high band (Table 4), each a band's own significance. Treating the frequencies as independent would give $+16.4$, $+12.3$, $+9.8$ and $+2.7\,\sigma$, an overcount. With McKay et al.'s sky (McKay et al., 2026) in place of GSM2016 it is smaller in every band, with a worst channel of $+4.0\,\sigma$ at 62.50 MHz: McKay et al.'s sky is brighter, and θ refitted on it is 0.79–0.84 times the GSM2016 value.

What the tension means: θ/G, about 1.62, is the noise source's kelvin scale times the sky-times-beam scale s ($\theta = kGs$). If GSM2016 and the full-sphere beam normalisation are right ($s = 1$), the field's kelvin scale is 1.62, against the bench's 1.15. The bench's scale error is one number for all frequencies, so it moves every band together. The difference $\Delta k = \theta/G - k_{\mathrm{bench}}$ does not follow z's fall across the band: its band means are 0.81 ± 0.13, 0.45 ± 0.12, 0.38 ± 0.13 and 0.56 ± 0.29. The top band's lower z comes from its larger error: it has 2 channels, and θ's error reaches 17 % at 226.56 MHz.

If the bench k is right, the antenna-side gain is $G_{\mathrm{bench}} = \theta/k_{\mathrm{bench}}$, 0.91–1.23, against the loss prior's median of 0.47–0.77. $G_{\mathrm{bench}} \geq 1$ at all 18 channels from 66.41 to 144.53 MHz (largest 1.23 at 101.56 MHz) and at 5 channels between 183.59 and 234.38 MHz, but there it is at most $1.8\,\sigma$ above 1, with σ from the bench's scale error and θ's jackknife error: G_{bench} is consistent with a lossless path. A passive coax and balun, however, cap G at the measured switch path's G_{sw} (0.78–0.91; Table 2), not at 1, and G_{bench} lies above G_{sw} at all 34 channels with a bench k, by 0.3–$2.5\,\sigma$. The conflict with the loss prior has the significance of Table 4. In $(\ln G, \ln k)$ the correlation is -0.08 to -0.99, strongest at 50–88 MHz (band mean -0.92), where the G prior is widest.

Against the previous prior. Under the prior the adopted one replaces (this memo's numbers on the same night fit and bench k with the previous prior, `inputs/previous_prior/`), $k = \theta/G$ was 1.84 ± 0.12, the band tensions were $+4.3$, $+4.0$, $+3.3$ and $+2.4\,\sigma$ ($+4.3$, $+3.9$, $+3.3$ and $+2.2\,\sigma$ with the previous prior on the night fit's θ, so the prior, not the load term's change of θ, moves them most), G_{bench} was 1.07, 1.05, 0.96 and 1.08, the sky-times-beam scale s of §4 was 1.98, 1.57, 1.51 and 1.79, and the load term moved θ or P beyond σ_J at 16 frequencies (now 4). The evidence-based prior moves G toward the bench but narrows it, so the low-band tension rises and the upper bands' fall. The bench still needs $G \approx 1$, which the parts at their datasheet values cannot supply: the k–G difference lies in the sky scale, the beam normalisation or the bench k (with its transfer to the field), unless the parts are far off their datasheets, which the in-situ check cannot yet test.

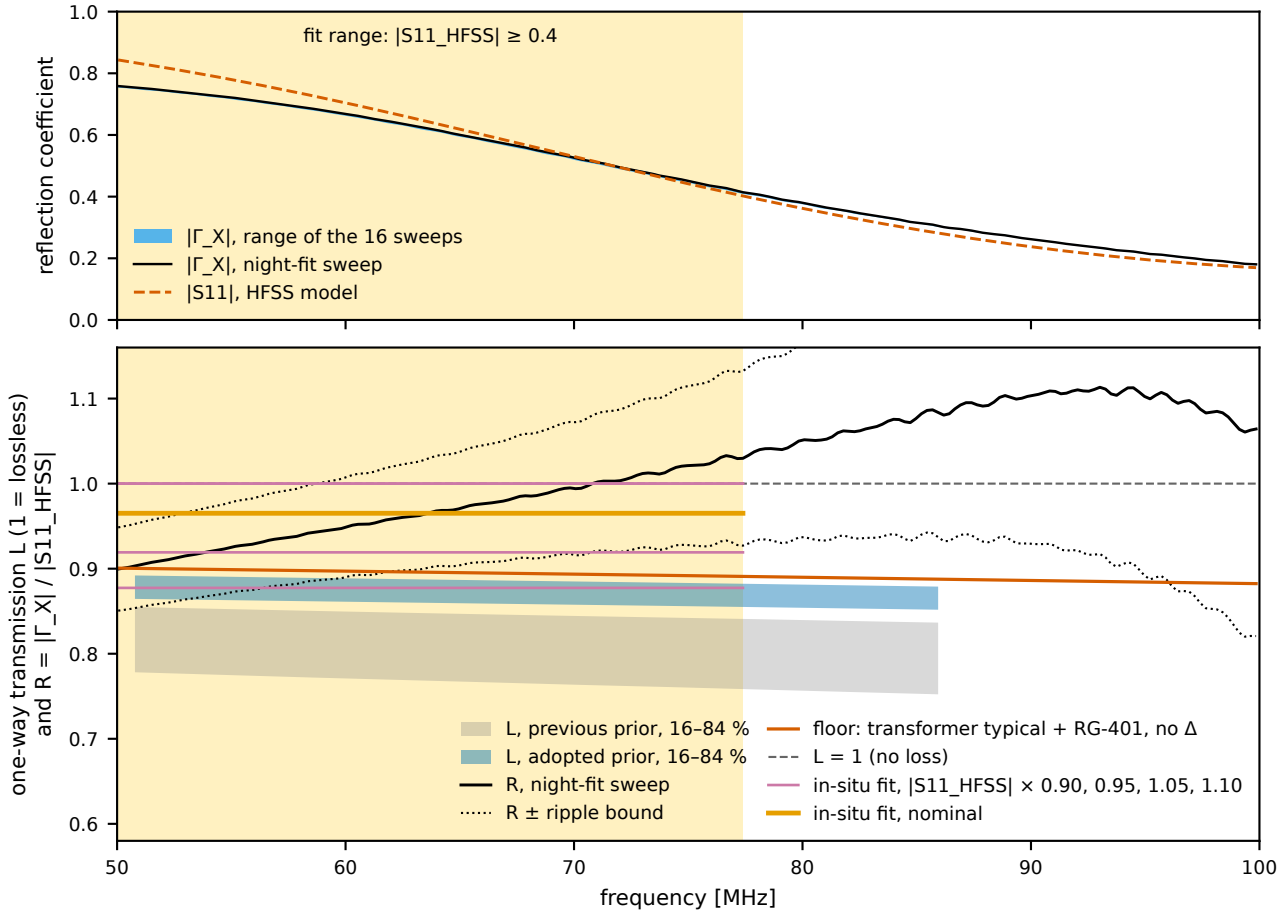


Figure 4: **The in-situ check of the loss, and why it is not yet usable.** 50–100 MHz; shaded: the fit range, $|S_{11,\mathrm{HFSS}}| \geq 0.4$. Upper: the measured reflection behind the switch's antenna port, $|\Gamma_X|$ (`tcal@v0004`'s antenna sweeps de-embedded through the measured switch path; black, the night fit's sweep; blue, the range of all 16 sweeps, mostly hidden by the black line), against the HFSS model's $|S_{11}|$ (dashed). Lower: $R = |\Gamma_X|/|S_{11,\mathrm{HFSS}}|$ for the night fit's sweep (black) with the bound on its ripple from the line's mismatch (dotted); the one-way transmission L (1 for a lossless line; lower is lossier) of the adopted (blue) and previous (grey) priors, 16–84 %; the datasheet floor (vermillion: the transformer's typical loss plus the coax at its typical loss, no allowance); $L = 1$ (dashed); and the in-situ fit of a passive line's loss to R (orange) and with HFSS's $|S_{11}|$ scaled by 0.90, 0.95, 1.05 and 1.10 (pink; the two lowest scalings reach $L = 1$). R rises across the fit range and exceeds the floor and 1, which no passive line does: the model's $|S_{11}|$, not the loss, sets its shape. Diagnostic.

Table 4: k against G by band. z: the tension of the band's weighted mean of Δk, with draws joint across frequency, and the worst channel, for GSM2016; z_{McKay}: the same with McKay et al.'s sky. $G_{\mathrm{bench}} = \theta/k_{\mathrm{bench}}$, G_{prior} (the prior's median) and ρ: band means, G_{prior} weighted by the prior's 16–84 % width. n: channels with a bench k.

	50–88	108–150	150–200	200–235 MHz
z (σ)	+6.0	+3.7	+3.0	+1.9
worst z (σ)	+6.4	+4.3	+4.3	+2.0
at (MHz)	70.31	109.38	191.41	226.56
z_{McKay} (σ)	+3.2	+1.5	+0.9	+0.7
G_{bench}	1.07	1.05	0.96	1.04
G_{prior}	0.66	0.76	0.73	0.70
$\rho(\ln G, \ln k)$	−0.92	−0.70	−0.61	−0.16
n	8	11	11	2

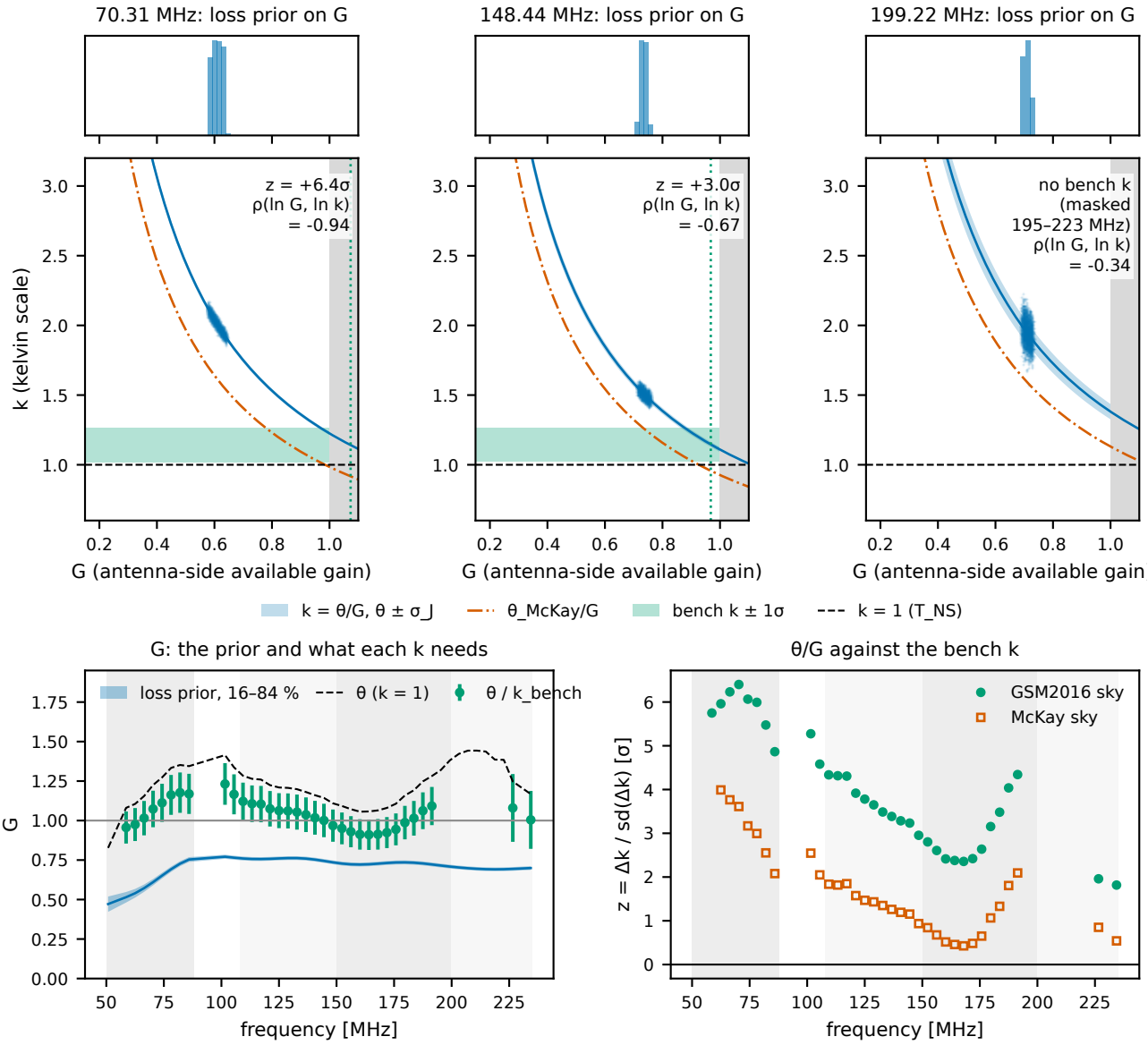


Figure 5: k **against** G**.** Upper panels, at 70.31, 148.44 and 199.22 MHz: the loss prior's draws of the antenna-side available gain G (marginal above) and, for each, $k = \theta/G$ with θ from the night fit $\pm\sigma_J$ (blue points; blue band: the hyperbola $kG = \theta$). Green: the bench k in `tcal@v0004`'s frame ±10.4–10.8 % (masked at 195.31–222.66 MHz), with the G it needs, θ/k_{bench} (dotted). Black dashed: $k = 1$, the noise-source temperature `tcal@v0004` carries. Vermillion dash-dotted: θ refitted on McKay et al.'s sky. Grey: $G > 1$, unphysical for a passive path. Lower left: the G prior against frequency (16–84 %) and the G that the bench k (green, $\pm1\sigma$; the bars reach $G = 1$) or $k = 1$ (black) would need. Lower right: the tension z for GSM2016 and McKay skies. Diagnostic.

Measured: θ (the night), k (the bench) and G_{sw}, independently. **Prior:** G_{cb}, from the parts' datasheets and the allowance Δ. **Breakers:** the coax and balun measured (a lab measurement before shipping, or the in-situ check with a validated antenna model); the sky scale (GSM2016 against McKay et al.); the beam normalisation (the raster's coverage); a field measurement of the noise source's temperature. **Status:** in tension. The sky scale, the beam normalisation and the bench k can take up the difference; the night fit cannot say which. The 200–235 MHz band rests on 2 channels, because of the bench mask.

Table 5: Noise waves from the D5 field calibrators for three noise-source temperatures (four-parameter model; means over its 682 channels). Columns 2–5: T_{NS} (K), band means over 50–88, 108–150, 150–200 and 200–237 MHz (the noise-wave fit's top band).

T_{NS} case	50–88	108–150	150–200	200–237	T_{unc} (K)	T_{cos} (K)	T_{sin} (K)	T_0 (K)
nameplate ENR	816	816	816	816	−205.6	−120.1	+98.7	+367.6
`noise_source@v0001`	582	521	498	489	−40.6	−49.5	+36.4	+139.3
bench	508	451	422	449	+0.6	−30.9	+21.4	+85.5

3.3 T_{gnd} against T_{off}: inside P

Over one parked night f is nearly constant, so for any T_{gnd} there is a T_{off} that gives the same P (Eq. 3), along a line of slope $-\theta\bar{f}$, so a ground temperature chosen by a prior is not a D5 measurement. The night fit splits P with the priors $T_{\mathrm{gnd}} \sim \mathcal{N}(293, 30)$ K and $T_{\mathrm{out}} \sim \mathcal{N}(290, 15)$ K (in T_{add}) and the loss prior on G (through k). The T_{gnd} prior comes from the site's soil temperature (memo 008, § 5.2): its centre lies between the soil a metre down in July and the deep annual mean, because the depth the emission comes from is not known, and its width covers rock against soil. It is a prior on the physical temperature: the ground's brightness, $(1 - R)\,T_{\mathrm{gnd}}$ plus the sky it reflects, is not covered by its width.

Figure 6 shows the split. The slope $-\theta\bar{f}$ is −0.79, −0.62 and −0.74 at 70.31, 148.44 and 199.22 MHz. Holding each input at its centre shows which one sets T_{off}: the T_{gnd} prior sets 58, 96, 95 and 79 % of T_{off}'s variance from low to high band (band means). The loss prior sets 20 % at 50–88 MHz (94 % at 50.78 MHz, where the antenna's mismatch widens G_{cb}; Table 2) and below 1 % above 108 MHz. The implied T_{off} (16–84 %) is −38 to 17, −3 to 34 and −2 to 43 K at the three frequencies. Moving the T_{gnd} prior's centre to 253 or 333 K moves T_{off} to 21 or −42 K at 70.31 MHz, 41 or −9 K at 148.44 MHz, and 51 or −8 K at 199.22 MHz.

Read the other way, setting $T_{\mathrm{off}} = 0$ gives the ground temperature the night's level implies under the model's black ground ($R = 0$; Fig. 6, bottom): 249–367 K from 66 to 235 MHz, rising to 378–481 K below 63 MHz (no crossing at 50.78 MHz). That is the size expected for the site, but a physical temperature has no structure in frequency, and this one varies by tens of kelvin across the band. It is a consistency check, not a measurement: T_{off} carries the noise-wave term kD and an unidentified remainder (§3.4), and a ground with $R > 0$ (emissivity below one, and the sky it reflects) changes the reading.

Measured: P. **Priors:** T_{gnd}, G, T_{out}. **Breakers:** a ground-fraction lever (calibrated data at a second height, or rotation through many sidereal times); T_{gnd} measured; T_{off} constrained by the receiver's noise waves (§3.4). **Status:** prior-determined. A T_{off} or a T_{gnd} from this night is reported as the prior's.

3.4 The noise waves against P: a candidate breaker, not adopted

Through the antenna's reflection coefficient, the receiver's noise waves enter P as k times the difference D between the antenna's and the load's noise-wave terms (Monsalve et al., 2017). `tcal@v0004` removes no noise-wave term, and the receiver's own T_0 cancels in its three-state calibration. T_{off} carries kD (Eq. 4) and any other additive term. A fit to the D5 field calibrators solves for the noise waves (the ambient load, the noise source, and the SP1 open and short) with the noise-source temperature fixed in three ways: the nameplate ENR, the team's `noise_source@v0001`, and the bench measurement. Table 5 gives the result of a four-parameter model (T_{unc}, T_{cos}, T_{sin} and T_0 per channel) for each.

Figure 7 shows the solutions against frequency. With the bench T_{NS}, T_{unc} averages +0.6 K, against −205.6 K with the ENR, but the bench T_{NS} does not pin it: up to 221.9 MHz, T_{unc} is −14 to +21 K, inside its total error of 22–33 K at every fitted channel. Above that, T_0 is negative at every fitted channel (−57 K at 229.0 MHz), which a receiver's noise temperature cannot be. None of the 12 candidates (3 T_{NS} cases, 4 models) passes the held-out test (0 pass). The test was fixed before any candidate was scored: each solution predicts calibration cycles and an SP1 termination left out of its fit, and is scored against baseline models. The held-out residual of the SP1 terminations has an rms of 57 K (short) and 114 K (open), against 6.9 and 6.0 K from the reflection coefficients' covariance: 8 and 19 times larger. It is odd in the termination: in delay, $(O - S)/2$ peaks at 34 K at 16.5 ns, against 9 K for $(O + S)/2$ at 17.0 ns. That is consistent with a $T_{\mathrm{cos}}/T_{\mathrm{sin}}$-like term or a phase error in Γ; the peak's coincidence with the cable's one-way delay (16.1 ns) identifies nothing.

The noise-wave prior (bench case) is that solution, with the band means and errors of Table 6. Its statistical covariance is scaled by the fit's χ^2 per degree of freedom (28.5) and adds 200 refits over reflection-coefficient

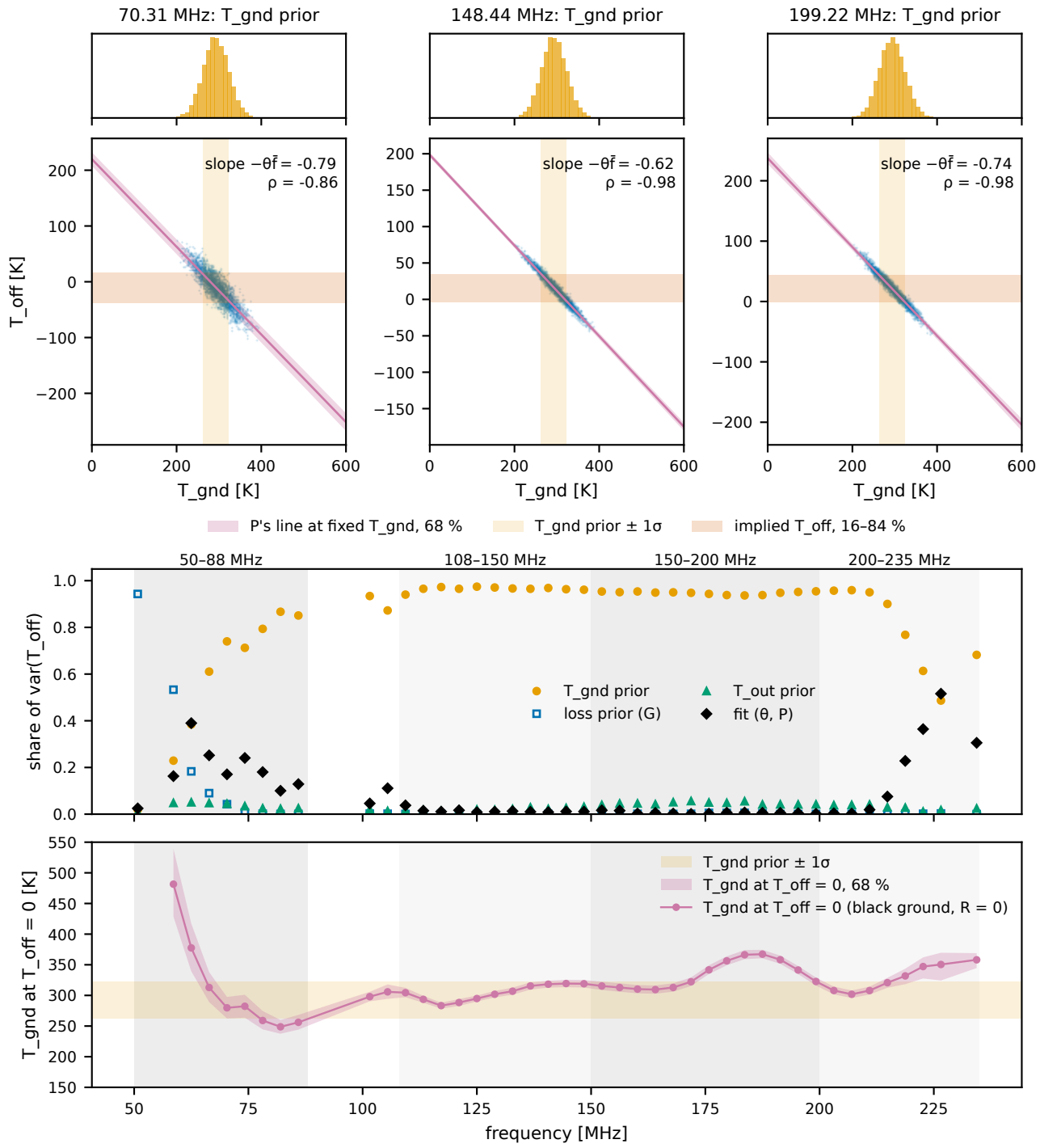


Figure 6: T_{gnd} **against** T_{off} **inside** P**.** Upper panels, at 70.31, 148.44 and 199.22 MHz: the night fit's split of P (Eq. 3; its draws, with the run's recorded seeds). The T_{gnd} prior $\mathcal{N}(293, 30)$ K is the marginal above and the yellow band. Blue points: each draw's T_{off}. Purple: the same line with T_{gnd} held at each value from 0 to 600 K (68 % over the other draws), slope $-\theta\bar{f}$. Pale vermillion band: the T_{off} this prior implies (16–84 %). Middle: the share of T_{off}'s variance removed by holding one input at its centre, against frequency. Shares need not sum to one because $k = \theta/G$ is nonlinear. Bottom: the T_{gnd} at which P's line gives $T_{\mathrm{off}} = 0$, under the model's black ground ($R = 0$), against frequency (purple: the median line's crossing, band: its 68 %), with the T_{gnd} prior (yellow). Diagnostic.

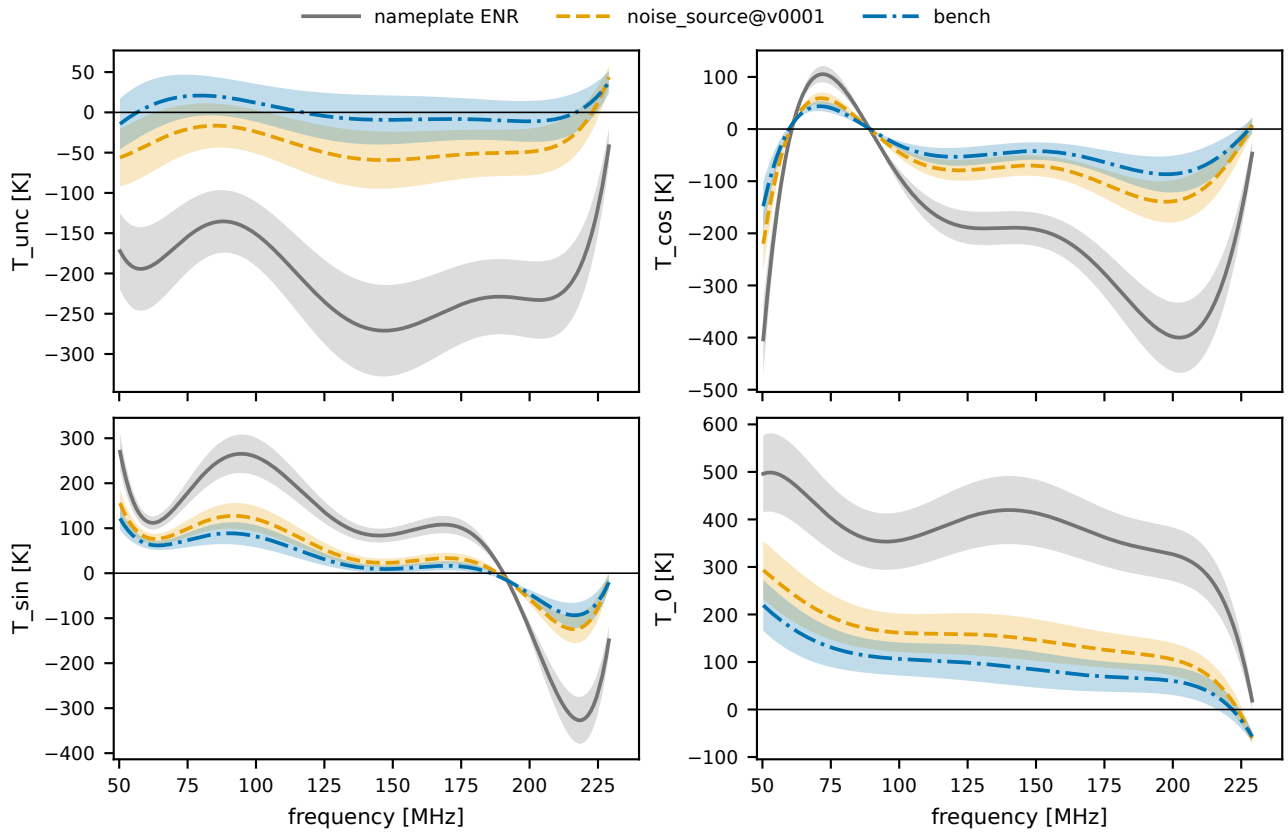


Figure 7: **The receiver's noise waves from the D5 field calibrators, for three noise-source temperatures.** The four-parameter solution ($T_{\rm unc}$, $T_{\rm cos}$, $T_{\rm sin}$, T_0) per channel over its fit channels, with the noise-source temperature from the nameplate ENR (grey, solid), the team's `noise_source@v0001` (orange, dashed) and the bench measurement (blue, dash-dotted); lines: mean, bands: ± the total error. With the bench temperature $T_{\rm unc}$ is consistent with zero up to 221.9 MHz and T_0 turns negative above it; the phase waves $T_{\rm cos}$ and $T_{\rm sin}$, which carry the shift of P in §3.4, depend on the case. None of these solutions passes the held-out test. Diagnostic.

Table 6: The noise-wave prior (bench $T_{\rm NS}$, four-parameter model): band mean ± total error, in kelvin.

Band (MHz)	$T_{\rm unc}$	$T_{\rm cos}$	$T_{\rm sin}$	T_0
50–88	$+12 \pm 29$	$+6 \pm 2$	$+77 \pm 16$	$+152 \pm 45$
108–150	-5 ± 29	-48 ± 17	$+29 \pm 13$	$+96 \pm 38$
150–200	-9 ± 27	-64 ± 24	$+3 \pm 6$	$+71 \pm 33$
200–237	$+14 \pm 19$	-37 ± 23	-47 ± 18	-3 ± 20

draws. Its caveats: the SP1 epoch, on 07-18 after the battery swap, is a different receiver state from the 07-17 night; $T_{\rm NS}$'s dependence on the diode-board temperature is unmeasured; and because the held-out test fails, the covariance understates the error. Its T_0 does not reach the night's P.

What the prior would do to P. Mapping this prior through the night's antenna reflection coefficient gives kD, with $k = \theta/G$ and a 20 % coherent amplitude error for the transfer from the SP1 epoch to the night (Fig. 8). D comes from $T_{\rm cos}$ and $T_{\rm sin}$: $T_{\rm unc}$'s share is at most 0.15 at any frequency. Taking kD out of $T_{\rm off}$ moves $T_{\rm off}$'s median by more than its own 16–84 % half-width at 10 of the 36 clean frequencies (66.4–214.8 MHz; the 10 within 66.4–85.9 and 175.8–203.1 MHz), and at 0 with the phase noise waves dropped (the even-only variant, V-even). Over each band's clean channels (6 at 66–86, 11 at 109–148, 13 at 152–199 and 4 at 203–215 MHz), the band-mean shift it predicts in P is $+23.3 \pm 5.3$, $+1.4 \pm 0.3$, -4.8 ± 2.5 and $+12.3 \pm 6.1$ K, against P's joint error of ±11.3, 4.0, 3.3 and 7.3 K: larger in 3 of the four bands, by 1.5–2.1 times. These errors come from the noise-wave

fit's covariance, which understates the error (above); with it inflated by α^2 (V-infl, Fig. 8) they are ±29, 1, 28 and 67 K. Per channel, the largest clean kD is −72 K at 66.41 MHz; below the clean range kD reaches −536 K (50.78 MHz), and those channels are not used. The noise-wave fit used the bench T_{NS}, which corresponds to $k \approx 1.15$, while $k = \theta/G$ is 1.46–2.10; with the bench k in place of θ/G, the self-consistent choice, the shift still exceeds P's error in 2 of the 3 bands with a bench k, and with the noise waves solved at `noise_source@v0001`'s temperature and $k = 1$ in 3 of four. With the phase noise waves dropped the shift is $+5.1 \pm 37.9$, -0.1 ± 1.3, -0.2 ± 7.1 and -0.6 ± 22.7 K, at most 0.45 of P's joint error in any band, so the shift comes from $T_{\cos}$ and $T_{\sin}$.

The data do not support the term. The held-out test of the SP1 terminations fails with a termination-odd residual (above). The shape test is the direct evidence (Fig. 8b). Across the 36 clean frequencies at 65–215 MHz, it fits the frequency structure of T_{off} (the split of P at the priors' centres) that a 3-term Chebyshev series leaves as w times the same structure of the predicted kD, each frequency weighted by T_{off}'s jackknife error, which also sets the σ of this χ^2. The prior predicts $w = 1$; the fit gives $w = -0.10 \pm 0.07$ (block jackknife), consistent with 0 and excluding 1: χ^2 is 2712 without kD ($w = 0$), 13774 at $w = 1$ and 2620 at the fitted w (3 % below $w = 0$, and still 73 per frequency): P does not carry the frequency structure the noise-wave prior predicts. The same projection on θ is the confounder test: a beam or sky ripple moves θ and P together along $\Delta P = -\bar{A}\,\Delta\theta$, so it would give w_θ like w. It gives $w_\theta = -0.90 \pm 0.11$, which does not match, so the test does not attribute P's remaining structure to the beam or the sky (it neither confirms nor excludes them). With the phase noise waves dropped, $w = -1.8 \pm 1.4$ and χ^2 at $w = 1$ is 2877.

If every other term of P were known ($T_{\mathrm{rest}} = 0$), adding kD would narrow θ's 16–84 % width by at most 3.6 %, at 4 of the 36 clean frequencies; a narrowing that holds only for an assumed T_{rest} is not a measured constraint, so no θ or k is taken from it.

So the noise-wave prior is a candidate breaker of the T_{gnd}–T_{off} split, not adopted: it moves T_{off} beyond its spread only through the phase noise waves, which the held-out residual and the shape test do not support, and it does not narrow θ without an assumed T_{rest}. It is material all the same: an unresolved systematic in P, larger than P's quoted error in 3 of four bands and not included in it. θ is not affected. The planned resolution is a bench measurement of the SP1 cable (all four S-parameters, repeated matings, several temperatures), followed by a refit of the noise waves with it.

Status: a candidate breaker, not adopted, and not in the night fit (open question 1, §7).

3.5 The other groups

Beam against sky. Beam-pattern errors and sky-map errors both change the shape of $S(t)$ and are degenerate on one parked night. The transmitter raster measures the beam independently of the sky (memo 005; open question 2, §7).

Tilt against a time trend. The night fit holds the box at $\tau = 0.19°$, the IMU's mean over the night from the readings before the firmware calibration's accelerometer bias was removed (the IMU's bins scatter about it by 0.54°, weighted); memo 003 (§ 7) gives the mean with the bias removed, and the fit moves to it at its next rerun. Against the 4.74° fitted with `empirical_raster_v0017` (memo 003), the measured tilt lowers the residual from 5.6–18.9 to 2.5–12.4 σ_{tot} (memo 008, § 5.1), and the bins in which the IMU shows the box moving are cut (§2). Broken for the 07-17 night.

Load term against the scale. $(1 - k)\, c_L(t)$ ties the load term to θ through k; it is a fixed column, iterated, and k is not read from it. Its k uses the loss prior's median G, so θ and P depend on the loss prior through it; it moves them by more than σ_J at 4 frequencies at a held-out cost of 27.17 to 27.50 (§2). Handled, with that dependence.

Gain drift against the sky. A time-varying gain against the sky's time dependence; the switch cycles give a measured gain series (open question 3, §7).

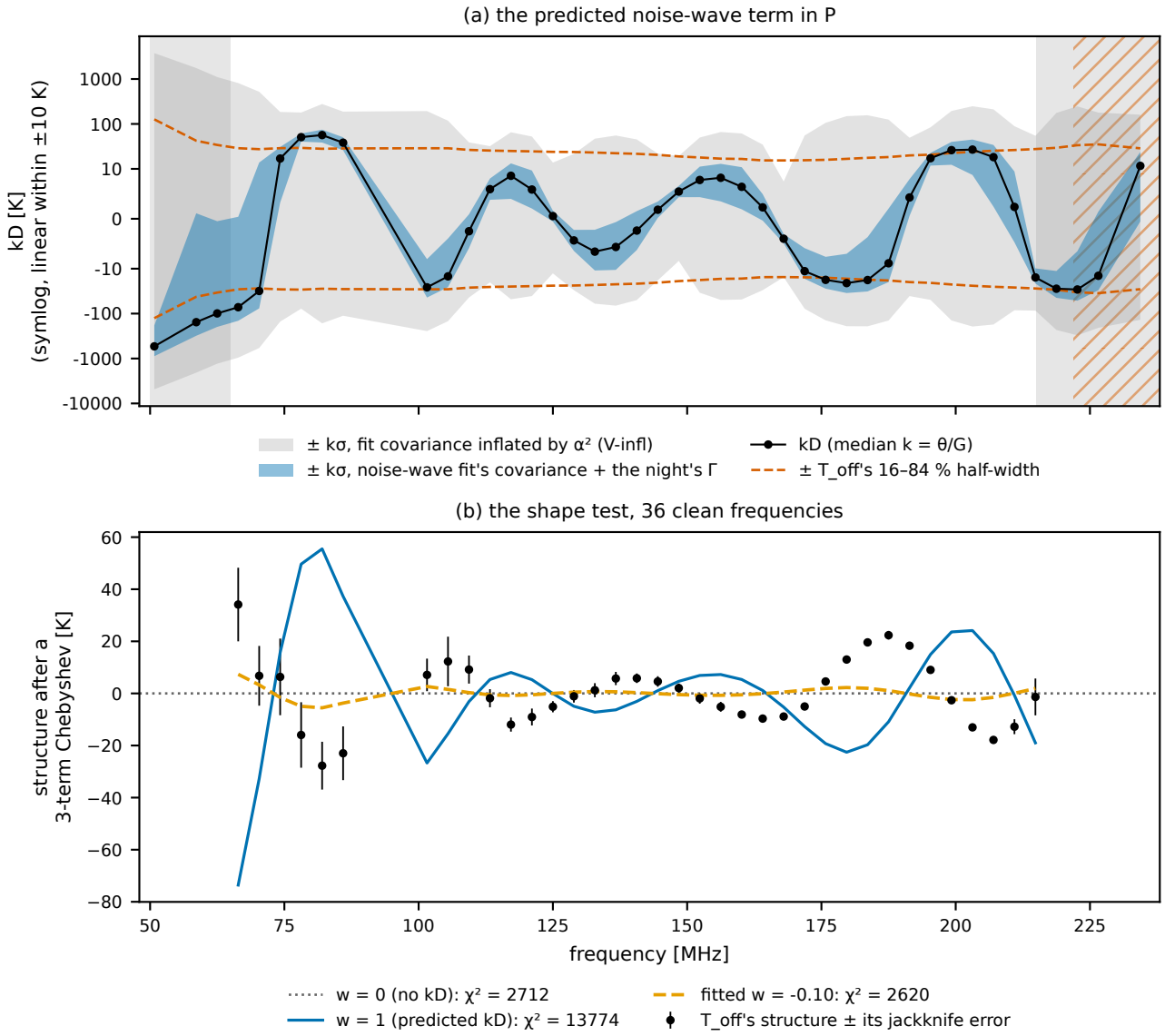


Figure 8: **The noise-wave prior predicts a term in P that the data do not carry.** (a) kD on the 07-17 night (black; the bench-case prior at the night's antenna reflection coefficient, k the median of θ/G), with $\pm k\sigma$ from the noise-wave fit's covariance and the night's reflection-coefficient draws (blue) and with the fit's covariance inflated by α^2, $\alpha = 12.2$ its largest held-out residual in units of σ (grey, V-infl). Vermillion dashed: $\pm$ the 16–84 % half-width of the night fit's $T_{\rm off}$; kD exceeds it at some frequencies within 66.4–85.9 and 175.8–203.1 MHz. Grey: outside the clean range (66.4–214.8 MHz); hatched: above 221.9 MHz, where the bench-case T_0 is negative. Vertical axis symlog, linear within $\pm$10 K. (b) The shape test over the 36 clean frequencies at 65–215 MHz: $T_{\rm off}$'s frequency structure after a 3-term Chebyshev series is removed (black, $\pm$ its jackknife error), against the predicted kD's structure at $w = 1$ (blue), at the fitted $w = -0.10$ (orange dashed) and at $w = 0$ (dotted), with each χ^2 in the legend. $T_{\rm off}$ does not follow the predicted ripple. The curves are recomputed from the stored inputs and reproduce the stored w and χ^2. Diagnostic.

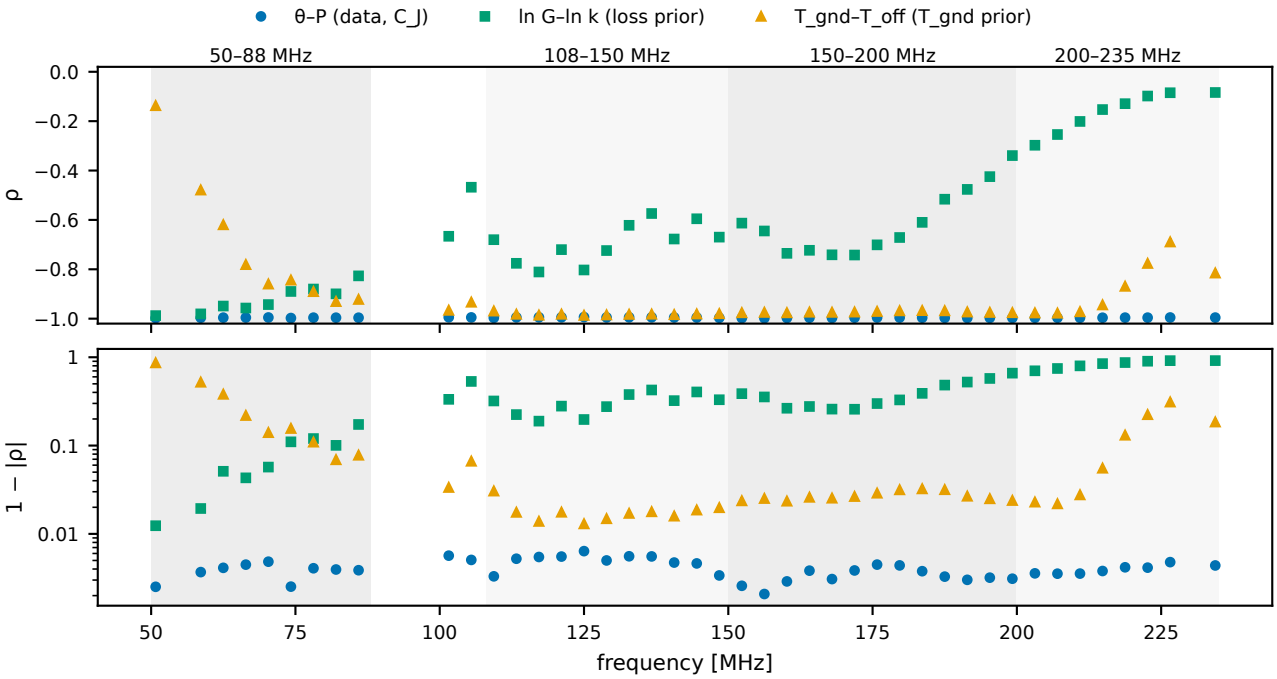


Figure 9: **The three pairs' correlation against frequency.** θ against P from the night fit's jackknife covariance (data only); $\ln G$ against $\ln k$ from the loss prior's draws and θ's jackknife error; $T_{\rm gnd}$ against $T_{\rm off}$ from the split of P with the $T_{\rm gnd}$, $T_{\rm out}$ and loss priors. Each pair is correlated in the coordinates its group is linear in. Upper: ρ; lower: $1-|\rho|$ (log), which separates the near-exact cases. The θ–P correlation is near-exact everywhere because P is the level at $A = 0$, far from the night's A; it is not a degeneracy of the data. The other two are as strong as their priors are wide. Grey bands: the four analysis bands (88–108 MHz is in none). Diagnostic.

4 What D5 can separate now, and under which conditions

Without conditions. The night measures θ and P jointly, and the mean level $M = \theta\bar{A} + P$ to 0.2–3.0 K across the band (§3.1). The IMU fixes the night's tilt.

If the bench k is right:

- With GSM2016's scale and the full-sphere beam normalisation, $G = \theta/k_{\rm bench}$ is 1.07, 1.05, 0.96 and 1.04 by band, against the loss prior's 0.66, 0.76, 0.73 and 0.70. Where $G_{\rm bench} \geq 1$ it is at most $1.8\,\sigma$ above 1, so a lossless path is allowed; but it lies above the measured switch path's own gain, and the parts' datasheets cannot supply it (§3.2.1, Fig. 3). Unless the coax and balun are far off their datasheets, at least one of two things gives: the sky is brighter than GSM2016 (McKay et al.'s sky is, and the tension drops to +3.2, +1.5, +0.9 and $+0.7\,\sigma$), or the beam's sky fraction is larger than the full-sphere normalisation gives. With the loss prior on G instead, the two together imply a sky-times-beam scale $s = \theta/(k_{\rm bench}\,G)$ of 1.72±0.20, 1.41±0.15, 1.33±0.15 and 1.50±0.28 by band (weighted means; the errors keep the bench's common 10.4–10.8 % scale instead of averaging it down), worst 1.9±0.2 at 58.59 MHz (Fig. 10, upper). On McKay et al.'s sky it is 1.37, 1.16, 1.11 and 1.19. $s = 1$ is below the 16th percentile at 34 of the 34 channels with a bench k (23 of 33 on McKay's sky). So if the bench k and the loss prior both hold, the sky, or the beam's response to it, is 1.3–1.7 times what GSM2016 and the full-sphere normalisation give. The two sky models change θ by 16–21 %, which closes at most 68 % of $s - 1$ in any band: on McKay et al.'s sky s is still 1.1–1.4, which needs a beam sky-fraction error of that size or parts far off their datasheets.
- The receiver's noise waves are not pinned by it: $T_{\rm unc}$ stays inside its error up to 221.9 MHz and T_0 is negative above (Table 5, Fig. 7), and the solutions fail the held-out test (§3.4), so the prior they give is diagnostic.

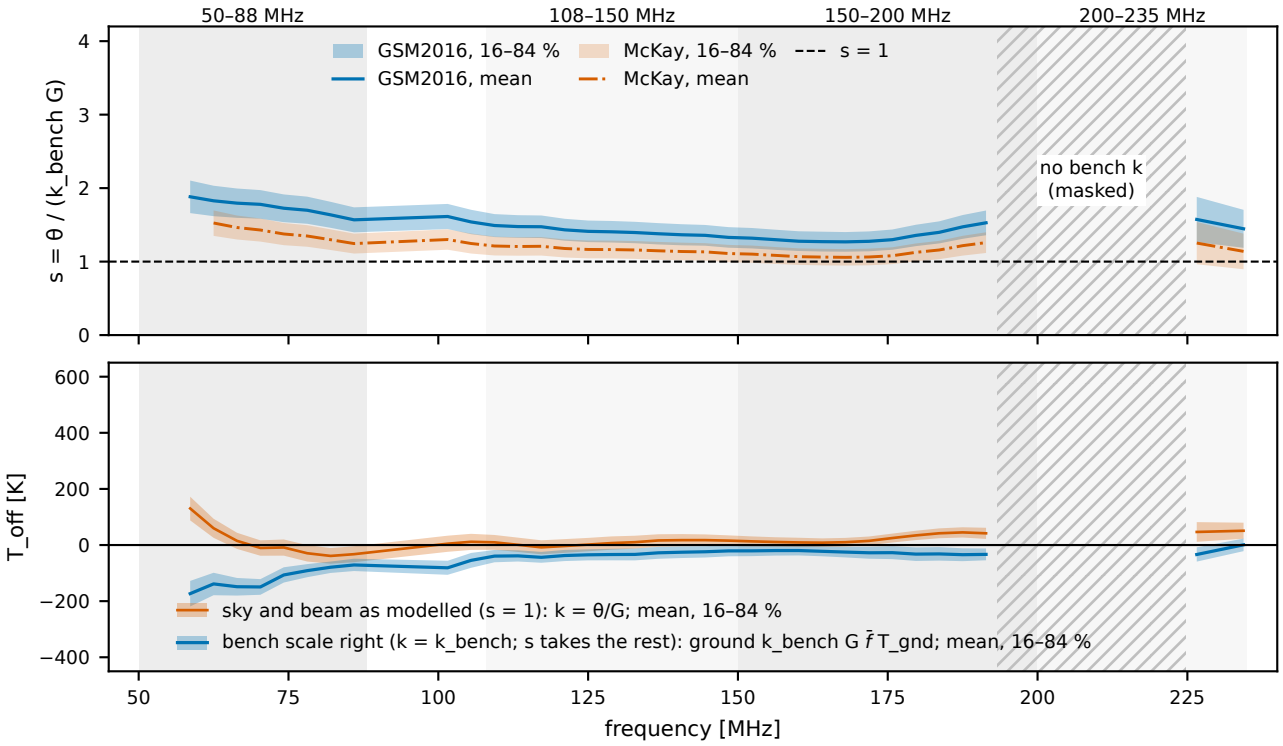


Figure 10: **If the bench k is right.** The night fit with the bench k as the kelvin scale (±10.4–10.8 %, one scale error common to all frequencies) and the loss prior on G; 10000 draws per frequency (stored means and 16–84 % ranges). Hatched: the bench mask (195.31–222.66 MHz; 50.78 MHz is masked too). Upper: the sky-times-beam scale $s = \theta/(k_{\mathrm{bench}}\, G)$ for GSM2016 (blue) and for θ refitted on McKay et al.'s sky (vermillion); $s = 1$ (dashed) if the sky scale, the beam normalisation, the bench k and the loss prior all hold. Lower: T_{off} at the two ends of the k–s trade-off. Vermillion: the sky and the beam as modelled ($s = 1$), so $k = \theta/G$ and the bench scale is the part that is off (as in Fig. 6). Blue: the bench scale right, $k = k_{\mathrm{bench}}$ with the ground term $k_{\mathrm{bench}} G \bar{f} T_{\mathrm{gnd}}$ of Eq. (3), so s takes the rest (the values quoted). Diagnostic.

- With $k = k_{\mathrm{bench}}$ in place of θ/G in the split of P (Eq. 3, with its ground term $k_{\mathrm{bench}} G \bar{f} T_{\mathrm{gnd}}$; the night fit's draws and priors), T_{off} is -105 ± 24, -33 ± 20, -26 ± 20 and -14 ± 24 K by band (worst -173 ± 46 K at 58.59 MHz), against $+0 \pm 28$, $+9 \pm 23$, $+22 \pm 17$ and $+49 \pm 31$ K with $k = \theta/G$ on the same channels (Fig. 10, lower, blue and vermillion). At $k = k_{\mathrm{bench}}$ the T_{gnd} prior sets 31, 65, 38 and 45 % of T_{off}'s variance by band, against 65, 96, 95 and 58 % at $k = \theta/G$. The ground term is never scaled by s: the beam is normalised over the full sphere, so s can only be the sky model's brightness or a beam-shape error acting on the sky. The 200–235 MHz band has 2 channels.

If the loss prior, GSM2016's scale and the beam normalisation are right, $k = 1.62 \pm 0.03$ (weighted mean), against the bench's 1.15: the bench measurement or its transfer to the field is then wrong.

If the T_{gnd} prior is right, T_{off} is -38 to 17, -3 to 34 and -2 to 43 K at 70.31, 148.44 and 199.22 MHz (16–84 %). Any other T_{gnd} moves it along the line of Fig. 6, so the night cannot test the condition.

5 What D6 must change

Each open breaker of §3 needs a change to the instrument, the lab preparation or the schedule:

- **The kelvin scale** k (§3.2: $\theta/G \approx 1.62$ against the bench's 1.15). A heated ambient load in the schedule, with spectra recorded while it is hot, so that T_{NS} is fitted in the field; a regulated, thermally stable noise-

source supply, with its voltage logged; thermistors on the SP1 cable, the balun–switch coax, the LNA and the noise-source pad; and a lab measurement of the ENR and the pad before shipping.

- **The antenna-side gain** G (§3.2.1: the loss prior's median G of 0.47–0.77 rests on datasheets and a judgement allowance, and the in-situ check is limited by the antenna model). S-parameters of every switch path, of the antenna-side coax and of the balun, before shipping; a connectorized, measurable balun, characterized before shipping; the switch paths measured in the assembled instrument; and lower loss and a better match ahead of the LNA.
- **The noise waves and** T_{off} (§3.4: the SP1 held-out residual is 8 and 19 times what the reflection coefficients' errors give). SP1 open and short and RFNOFF dwells all night, balanced, so that the four-parameter noise-wave fit can run per night; SP2 as a fourth calibrator, whose reflection coefficient does not lie on one circle in the complex plane with the others' (calibrators on one circle leave the four-parameter fit degenerate); a second high-$|\Gamma|$ standard on another switch path, which tests the reflection transfer behind the termination-odd residual; and the receiver's noise waves measured with external loads.
- T_{gnd} (§3.3: the prior sets 58–96 % of T_{off}'s variance, by band). The rock or soil temperature logged in the beam's ground footprint, and a full-azimuth spin at zenith.
- T_{gnd} **against** T_{off} (§3.3: f is nearly constant over a parked night). A change of the ground fraction during the deployment, with calibrated data at one kelvin scale while the horizon moves through the beam: a second observing height, or many orientations. The repeated rasters and the zenith spin also bear on it.
- **Beam against sky** (§3.5). Full rasters repeated across sidereal time, and a raster sized to the battery, with a repeat pass.

6 Caveats

- The night-fit residual is structured (memo 008, § 5.1); every split here is conditional on Eq. (1). Its cause is open.
- The bench k's transfer from bench to field is assumed flat in frequency (one scale error); the bench is masked at 195.31–222.66 MHz.
- Every temperature is in `tcal@v0004`'s frame; the next release scales temperature differences from the reference load's by 0.85–0.88 (§2).
- The tension z is the mean of the paired draws of Δk over their standard deviation. The draws of θ/G are skewed by the wide G prior, so z is not a tail probability.
- The loss prior rests on the transformer's typical insertion loss; the datasheet gives no maximum, so a unit far off typical is covered only by Δ, whose upper end is a judgement. The in-situ check cannot yet test it (§3.2.1).
- The McKay tension and the McKay s use the night fit's jackknife error, because no McKay jackknife is stored.
- Band errors come from draws joint across frequency (θ and P from the jackknife, one T_{gnd} and one T_{out}, one bench scale, the loss prior's templates); the per-frequency curves use draws independent per frequency.

7 Open questions

1. **The noise waves in** P**.** How much of T_{off} is kD, and is the phase part of the noise waves real (§3.4)? It needs a noise-wave solution that passes the held-out test, which the bench measurement of the SP1 cable is planned to give, and a bench-to-field transfer for the prior.
2. **Beam against sky.** Is the sky-times-beam scale s the sky model's brightness or a beam-shape error? The raster beam measures the beam independently of the sky; its posterior needs the fraction of each bin's beam-weighted sky the raster sampled.
3. **Gain drift.** Does a time-varying receiver gain take up part of the sky's variation over the night? The switch cycles give a measured gain series to test it against.

4. **The remainder $T_{\rm rest}$.** What additive term is left in $T_{\rm off}$ after kD? It has no model; until it is bounded, no split of P is a measurement.

5. **Which member of θ takes the k–G difference?** The parts' datasheets cannot supply the G the bench needs, so the sky's brightness, the beam's sky fraction or the bench k, unless the parts are far off their datasheets. Bench measurements of the coax and balun (or an antenna model good enough for the in-situ check), the raster beam and a field measurement of the noise source's temperature decide between them.

Revision history

2026-11-06 The inputs' metadata carry neutral provenance (product names, source commits, SHA-256); no number changes. The night fit's tilt, 0.19°, is described as the tilt the fit holds, the IMU's night mean before the accelerometer bias was removed (memo 003, § 7).

2026-11-05 First version. The night fit's degenerate groups on `tcal@v0004` as posteriors (θ–P, k–G, $T_{\rm gnd}$–$T_{\rm off}$), the evidence-based loss prior and its in-situ check, the noise-wave prior from the field calibrators and its map into P (a candidate breaker, not adopted), and what D6 must change.

References

McKay, L., Subrahmanyan, R., Chippendale, A. P., et al. 2026, Nature Astronomy, "Precise measurement of the absolute sky brightness at 60–350 MHz", doi:10.1038/s41550-026-02888-5, arXiv:2509.11846

Monsalve, R. A., Rogers, A. E. E., Bowman, J. D., & Mozdzen, T. J. 2017, ApJ, 835, 49, "Calibration of the EDGES High-band Receiver to Observe the Global 21 cm Signature from the Epoch of Reionization"

Zheng, H., Tegmark, M., Dillon, J. S., et al. 2017, MNRAS, 464, 3486, "An improved model of diffuse galactic radio emission from 10 MHz to 5 THz"

A Reproducing this memo

This memo's figures, tables and numbers come from `python make_products.py` in this directory, with numpy and matplotlib only, then `pdflatex memo-009.tex` twice. The script reads only the files in `inputs/`, checks each against its SHA-256 before use, and checks that they belong together (each downstream analysis read the night-fit run copied here, and every run started from a clean source). It simulates nothing and fits nothing beyond recomputing the shape test (below): every quoted number is a stored value, or a minimum, maximum, count or plain mean over a stored per-frequency curve. All ten figures are drawn from stored arrays; the shape test's curves (Fig. 8b) are recomputed from them and reproduce its stored w and χ^2 values.

The inputs are products of Christian Hellum Bye's EIGSEP D5 analysis of the 07-17 night, on `tcal@v0004` and `fit_v0025`: the night fit at the IMU's tilt (`inputs/night_fit/`), its degenerate posteriors (`inputs/posteriors/`), the noise-wave fit to the field calibrators (`inputs/noise_waves/`), its map into P and the shape test (`inputs/noise_wave_term/`), the loss prior's evidence and the in-situ check (`inputs/loss/`), the check that the prepared `tcal@v0005` build carries the bench $T_{\rm NS}$ (`inputs/tcal_v0005/`), and this memo's numbers from an earlier build under the previous loss prior (`inputs/previous_prior/`). Each analysis's `PROVENANCE.json` records its seeds, priors and parameters. `inputs/night_fit/extract.npz` holds the night fit's delete-one-block replicates and, at the three shown frequencies, the loss-prior and ground-temperature draws drawn in Figs. 5 and 6; `make_products.py --extract` regenerated them from the night fit's full products with the run's recorded seeds, and they reproduce its stored k and $T_{\rm off}$ percentiles exactly at every frequency.

input (`inputs/`)	content	SHA-256 (first 16)
`night_fit/results.json`	night fit: per-frequency results	`77f16cb3d9e422d0`
`night_fit/PROVENANCE.json`	night fit: run record (seeds, priors)	`e8404760c396e28c`
`night_fit/nightfit_numbers.json`	night fit: summary numbers	`e175ed8cc34fa868`
`night_fit/extract.npz`	night fit: replicates and draws (–extract)	`460ac65c0f52e7fc`
`posteriors/results.json`	degenerate posteriors: results	`a474bfdbf12bcd89`
`posteriors/PROVENANCE.json`	degenerate posteriors: run record	`e3523f998585649d`
`posteriors/curves.npz`	degenerate posteriors: curves	`40c627cc15db75ba`
`noise_waves/results.json`	noise-wave fit: results	`6dc2f7403c42890d`
`noise_waves/PROVENANCE.json`	noise-wave fit: run record	`3d89d8087e63b6aa`
`noise_waves/noise_wave_prior.json`	noise-wave prior: summary	`858f32234fa3d522`
`noise_waves/F_noise_waves_total.csv`	noise-wave fit: per channel	`056b7157bb4cbc59`
`noise_wave_term/results.json`	noise-wave term in P: results	`7abd335b303e3e0e`
`noise_wave_term/PROVENANCE.json`	noise-wave term in P: run record	`13b0c223a2e11a47`
`loss/results.json`	loss prior's evidence: results	`447e21f09c98b552`
`loss/PROVENANCE.json`	loss prior's evidence: run record	`fafe5d2c3697324b`
`loss/insitu_loss.npz`	in-situ loss check: arrays	`8a15054b59fd2caa`
`tcal_v0005/compare.json`	tcal v0005 build: T_{NS} check	`152ae57ae5be5b60`
`previous_prior/numbers.json`	previous loss prior: earlier build's numbers	`329c360c212cff9c`
products generated at	this repository	`ce766db`
numpy, matplotlib		2.3.5, 3.11.2